PYTHON ACTIVITY-BASED LEARNING

Case Study 07: Nested List Examination Seating Validation

Dr. RVR NRI Institute of Technology | Python Programming

Student Name: _____ **Roll No:** _____

Branch / Section: _____

Problem Statement

Validate a classroom seating matrix for duplicate roll numbers, reserved seats and empty seats.

Objectives

• Represent classroom seating using a nested list.

• Detect duplicate student roll numbers.

• Count empty and reserved seats.

Assumptions

• The value 'R' represents a reserved seat and 0 represents an empty seat.

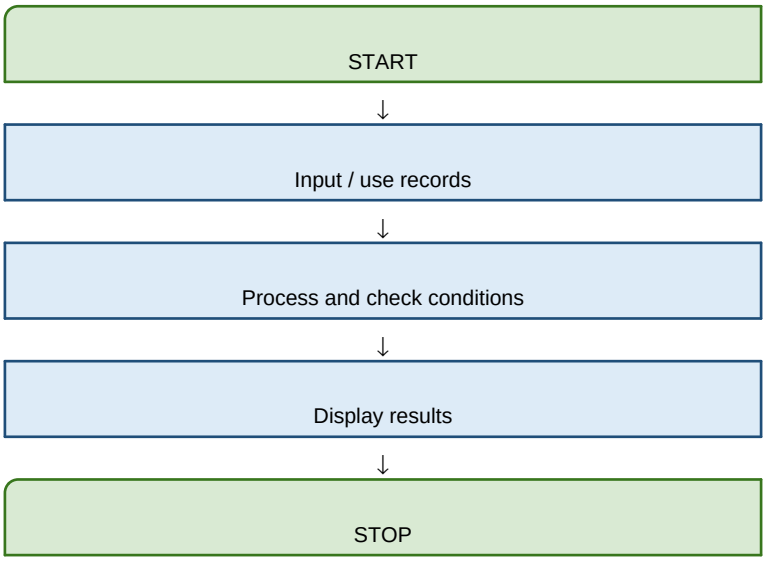
• Other values are student roll numbers.

• A roll number must appear only once.

Algorithm

1. Create a seating matrix as a nested list.
2. Traverse every row and seat using nested loops.
3. Skip reserved seats and count empty seats.
4. For student roll numbers, check whether the number has already appeared.
5. Report duplicates and display seat counts.

Flowchart



Pseudocode

```
START
STORE seating matrix
SET seen, empty, reserved
FOR each row and seat
    IF seat is 0, count empty
    ELSE IF seat is R, count reserved
    ELSE IF seat is in seen, report duplicate
    ELSE add seat to seen
DISPLAY counts
STOP
```

Python Source Code

```
seats = [
    [101, 102, "R", 104],
    [105, 102, 0, 108],
    [109, "R", 0, 111]
]

seen = []
empty = 0
reserved = 0

for row in seats:
    for seat in row:
        if seat == 0:
            empty += 1
        elif seat == "R":
            reserved += 1
        elif seat in seen:
            print("Duplicate roll number:", seat)
        else:
            seen.append(seat)

print("Occupied seats:", len(seen))
print("Reserved seats:", reserved)
print("Empty seats:", empty)
```

Test Cases

Test input / condition	Expected result
Roll number 102 appears twice	Duplicate 102 reported
Two values are 0	Empty seats = 2
Two values are R	Reserved seats = 2

Output Screenshot

Sample console output (formatted to show the expected execution result):

```
Duplicate roll number: 102
Occupied seats: 7
Reserved seats: 2
Empty seats: 2
```

Conclusion

The program checks seat allocation and counts occupied, reserved and vacant positions using nested lists.